Ramy Rady

Senior Mixed-Signal IC Design Engineer · Ph.D. in Silicon Photonics

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Summary

Senior mixed-signal IC designer with **4 years of industry experience** (2.5 years at Apple Silicon, 1.5 years at Fraunhofer IIS) and a **Ph.D. in silicon photonics** from Texas A&M, plus internships at Meta Reality Labs and Qualcomm. At Apple I own multiple high-speed SerDes IPs in **leading-edge CMOS** process nodes, from schematic through power-integrity closure and silicon, and build AI-assisted (Claude) design and verification automation. My Ph.D. covered silicon-photonic receivers, ring-resonator and APF filters with **automatic wavelength tuning**, and CMOS ↔ Si-photonics hybrid integration, with 10 IEEE/Optica papers (6 first-author) and 3 patent applications.

Work Experience

Apple Inc., Silicon Engineering Group — Cupertino, CA *SerDes Transceiver Analog-Mixed-Signal Design Engineer · May 2024 – Present*

- **Own multiple IPs** for next-generation high-speed SerDes transceivers in Apple Silicon SoCs, from transistor-level schematic through block and IP-level validation, signoff and silicon bring-up.
- Designed many **high-performance, high-speed analog circuits and custom digital cells** in leading-edge process nodes.
- Run **power-integrity** analysis and closure for owned IPs, working with layout to meet EM/IR and supply-noise targets.
- Built **AI-assisted design and verification automation with Claude**: scripted simulation setup, regressions and results checking, which makes design iteration faster and more repeatable.
- Lab characterization and model-to-silicon correlation on chips shipping in products; partner with PHY, packaging and system architects on PPA.

Texas A&M University — College Station, TX *Ph.D. Researcher — Silicon-Photonic / RF-CMOS Hybrid Systems · Jan 2019 – Apr 2024*

- Architect and lead designer of a **20–43.5 GHz wideband tunable silicon-photonic receiver front-end** for mm-wave channel selection / jammer rejection (*J. Lightwave Technol.* 2023).
- Built an end-to-end **automatic tuning system for silicon-photonic ring resonators and 4th-order APF pole/zero filters**, the same wavelength-locking problem DWDM CPO faces (RWW 2023 Best Student Paper Finalist; OFC 2021/2022; MWP 2022).
- Designed a closed-loop calibration platform (TIA + ADC + DAC + μC) and an **ML algorithm that dynamically adjusts pole locations** under PVT and thermal drift.
- Demonstrated a **CMOS / Si-photonics hybrid-integrated SDR receiver** with >80 dB rejection of <−10 dBm in-band blockers (IEEE RFIC 2023).
- Co-designed a 20–35 GHz MZM driver with programmable linearizer (IEEE TMTT 2023) and a **12.5 Gb/s 1.38 mW inverter-based optical receiver in 28 nm CMOS** (IEEE MWSCAS 2022).
- Authored 5 successful research proposals to federal and industrial funding agencies. Advisor: Prof. Kamran Entesari.

Meta Reality Labs — Redmond, WA *Research Scientist Intern — CMOS Drivers & Photonic Stabilization Loops · May 2023 – Aug 2023*

- Designed a 130 nm **electronics driver + TIA for a zonal-illumination photonic chip** used in next-gen AR/VR displays; improved efficiency and luminance control.
- Co-architected closed-loop **photonic bias stabilization** for a laser + MEMS subsystem under thermal / mechanical drift.

- **3 patent applications filed**: laser-driver topology, MEMS driver, auto-stabilization loop.

Qualcomm Inc. — Dallas, TX *Engineer Design Intern — mm-Wave IC R&D · May 2020 – Aug 2020*

- Designed mm-wave power amplifiers for 5G radio; built a PA lifetime / reliability calculator the group adopted for architecture reviews.

Fraunhofer IIS — Erlangen, Germany *R&D Engineer — CMOS RF / mm-Wave · May 2017 – Nov 2018*

- Co-designed and taped out a **10 GHz transceiver + PLL chip** in CMOS for automotive AR/HUD; the chip was licensed to **Bosch for a BMW product**.

Education

Jan 2019 – May 2024	**Ph.D., Electrical & Computer Engineering** — Texas A&M University, USA · GPA 3.71/4.0 Advisor: Prof. Kamran Entesari · Dissertation: *Silicon-Photonic / CMOS Hybrid Front-Ends for mm-Wave Channelization and Automatic Tuning*
Feb 2015 – Jun 2017	**M.Sc., Electrical & Computer Engineering** — Istanbul Şehir University, Turkey · GPA 4.0/4.0 Thesis: *Ultra-Low Power, Low-Voltage Transmitter at ISM Band for Short-Range Transceivers (SC-DC-DC supplied)*
Sep 2008 – Jun 2013	**B.Sc., Electronics & Communications Engineering** — Ain Shams University, Cairo, Egypt · Graduated with Honors

Technical Skills

EDA Tools	Cadence Virtuoso (Schematic / Layout / ADE, Verilog-A/AMS), Spectre
RF / EM Simulators	Sonnet, Ansys HFSS, ADS, Microwave Office
Programming	Python (NumPy / ML), C/C++, MATLAB, Arduino
Scripting & AI	TCL, Perl; AI-assisted (Claude) design and verification automation

Selected Publications

*First-author papers marked with name in **bold**.*

Journals

1. **R. Rady**, C. Madsen, S. Palermo, K. Entesari, "A 20–43.5 GHz Wideband Tunable Silicon-Photonic Receiver Front-End for mm-Wave Channel Selection / Jammer Rejection," *J. Lightwave Technol.*, 2023.
2. Y.-L. Luo, **R. Rady**, K. Entesari, S. Palermo, "A Power-Efficient 20–35 GHz MZM Driver with Programmable Linearizer for Analog Photonic Links in 28 nm CMOS," *IEEE TMTT*, 2023.
3. **R. Rady** et al., "An Ultra-Low-Power Integrated Radio TX Link Supplied from a Switched-Capacitor DC–DC Converter in 65 nm CMOS Achieving 2 Mbps," *IEEE TCAS-II*, 2020.

Conferences

1. **R. Rady**, C. K. Madsen, S. Palermo, K. Entesari, "Automatic Tuning of Microwave Silicon-Photonic Ring Resonators," *IEEE SiRF / RWW*, 2023 — *Best Student Paper Finalist*.
2. **R. Rady**, Y.-L. Luo, C. Madsen, S. Palermo, K. Entesari, "A mm-Wave CMOS / Si-Photonics Hybrid-Integrated Software-Defined Radio Receiver Achieving >80 dB Blocker Rejection of <−10 dBm In-Band Blockers," *IEEE RFIC*, 2023.
3. **R. Rady**, C. Madsen, S. Palermo, K. Entesari, "A Silicon-Photonics Automatically-Tunable mm-Wave Remote Antenna Unit," *IEEE MWP*, 2022.
4. **R. Rady**, G. Choo, C. Madsen, S. Palermo, K. Entesari, "External-Modulator-Based Automatic Tuning of Reconfigurable Silicon-Photonic 4th-Order APF Pole/Zero Filters," *OFC*, 2021.
5. K. Entesari, S. Palermo, C. Madsen, G. Choo, **R. Rady**, S. Cai, B. Wang, "Automated Tuning for Silicon-Photonic Filters," *OFC*, 2022 *(invited)*.
6. P. Yan, C. Hong, P.-H. Chang, H. Kang, D. Annabattuni, A. Kumar, Y.-H. Fan, R. Liu, **R. Rady**, S. Palermo, "A 12.5 Gb/s 1.38 mW Inverter-Based Optical Receiver in 28 nm CMOS," *IEEE MWSCAS*, 2022.

7. Y.-L. Luo, A. Ershadi, **R. Rady**, K. Entesari, S. Palermo, "A Power-Efficient 20–35 GHz MZM Driver with Programmable Linearizer in 28 nm CMOS," *OFC*, 2021.

Patent Applications (Meta Reality Labs, 2023)

- Laser-driver topology for zonal-illumination AR/VR displays
- MEMS driver for steerable photonic display elements
- Auto-stabilization loop for laser + MEMS photonic subsystem

Awards & Honors

RWW 2023	Best Student Paper Award Finalist
RFIC 2023	Best Three-Minute Thesis Finalist
ISSCC 2024	Student Travel Grant
TAMU 2019–2024	Graduate Research Scholarship · Quality Graduate Student Award · Graduate Assistant Lecturer (Teaching Fellowship)
IEEE MTT-S	Student Ambassador